RAKESH KUMAR

SOFTWARE DEVELOPMENT ENGINEER (SDE)

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CAREER OBJECTIVE

Highly motivated Computer Science Engineering postgraduate with strong expertise in **C++, Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP)**, and System Design principles. Passionate about solving complex algorithmic challenges, optimizing low-level performance, and building scalable software applications. Seeking an entry-level Software Development Engineer (SDE) role to deliver clean, efficient code and robust software solutions.

TECHNICAL SKILLS

Programming Languages:	C++ (Modern C++11/14/17), C, Python, Java
Core CS Concepts:	Data Structures & Algorithms (DSA), OOPs, Operating Systems, DBMS, Computer Networks
Databases:	SQL (MySQL / PostgreSQL), Neo4j (Graph DB)
Developer Tools:	Git, GitHub, VS Code, Linux/Bash, GDB Debugger, Valgrind, Google Colab
AI / ML Knowledge:	Machine Learning Basics, Deep Learning (CNN, BiLSTM, GNN)

SOFTWARE ENGINEERING & C++ / DSA PROJECTS

High-Performance Custom In-Memory Key-Value Store | C++, STL, Multithreading **C++ / System Project**

- Architected a thread-safe, high-concurrency in-memory Key-Value store using C++ STL and standard threading libraries (std::mutex, std::lock_guard).
- Implemented dynamic Hash Map with custom chaining for O(1) average lookup/insertion, and integrated an LRU (Least Recently Used) Cache using a Doubly Linked List for automatic memory eviction.
- Optimized memory allocation and pointer management using C++ Smart Pointers (std::unique_ptr, std::shared_ptr) to prevent memory leaks and dangling pointers.

Custom File System & Shell CLI | C++, Data Structures, OS Principles **C++ / Operating Systems**

- Developed a interactive command-line shell simulating Unix file system operations (mkdir, cd, ls, touch, rm, cat) using advanced Graph and N-ary Tree data structures.
- Designed file indexing using B-Trees and Trie algorithms for fast string search and autocomplete functionality across nested directories.
- Handled error handling, input parsing, and efficient memory management using OOP principles and C++ File I/O streams.

Robust Multi-Temporal Satellite Image Analysis | Python, Deep Learning, GNN **Academic AI/ML Project**

- Developed an end-to-end Machine Learning framework for environmental change monitoring using multi-temporal satellite imagery.
- Implemented deep learning architectures including CNN, BiLSTM, and Graph Neural Networks (GNN) to process spatial-temporal data accurately.

Java Distributed Group Chat Application | Java, Networking, OOP **Software Development**

- Built a multi-client real-time chat application using Socket Programming and Multithreading in Java.
- Applied core Object-Oriented Design patterns to manage user sessions, application logic, and concurrent message broadcasting cleanly.

EDUCATION

M.E. - Computer Science Engineering (AI & ML) **Currently Pursuing**

Chandigarh University, Gharuan

B.Tech - Computer Science Engineering **Graduated**

Dev Bhoomi Uttarakhand University

RELEVANT COURSEWORK

Data Structures & Algorithms • Object-Oriented Programming (OOP) • Operating Systems • Computer Networks • Database Management Systems (DBMS) • Software Engineering • System Design Basics